An Explainable AI-Driven Agricultural Decision System Using Computer Vision, Ensemble Learning, and Retrieval-Augmented Generation

Rahul Chand
Dept. of Computer Science and Engineering
Lovely Professional University
Punjab, India
rahulchand938@gmail.com

Pragya Chauhan
Dept. of Computer Science and Engineering
Lovely Professional University
Punjab, India
pragyac156@gmail.com

Gaurav Pathak
Dept. of Computer Science and Engineering
Lovely Professional University
Punjab, India
pathakgaurav21june@gmail.com

Devansh Mandhotra
Dept. of Computer Science and Engineering
Lovely Professional University
Punjab, India
devanshmandhotra@gmail.com

Karan Kumar Das
UpGrad Education Private Limited
Bangalore, Karnataka, India
ORCID: 0009-0005-0088-4306
karandas4643@gmail.com

***Abstract*—Indian agriculture faces two persistent challenges early crop disease identification and accurate market price forecasting. Existing digital tools address these problems in isolation, forcing farmers to navigate multiple disconnected platforms, a barrier particularly significant for smallholder farmers with limited technical literacy. This paper presents the Intelligent Agricultural Decision Support System (IADSS), a unified platform integrating plant disease detection and crop price forecasting under a single interface. Disease identification uses a convolutional neural network trained on leaf imagery for multi-class classification. Price forecasting employs an ensemble of Random Forest and Prophet models coordinated through a meta-learner, consistently outperforming either model individually. A Retrieval-Augmented Generation chatbot completes the system, grounding responses in retrieved agricultural knowledge and live model outputs to deliver explainable, actionable recommendations to non-technical users. Evaluation confirms the ensemble achieves lower error rates than individual baselines, while the disease module reaches reliable classification accuracy on held-out images. The work demonstrates that combining predictive models with a knowledge-grounded conversational layer produces a system capable of translating raw predictions into decisions farmers can act on.**

***Index Terms*—Farmers, Agriculture, Disease Detection, Forecasting, Chatbot, Recommendations, Ensemble Learning, Retrieval-Augmented Generation, Convolutional Neural Network**

I. Introduction

Agriculture constitutes the backbone of the Indian economy, employing approximately 58% of the total workforce and contributing nearly 17% to the national GDP [16]. Despite its scale, the sector continues to suffer from preventable losses caused by two persistent and interrelated problems: the late detection of crop diseases and the inability of farmers to anticipate market price movements with sufficient accuracy [18].

Crop diseases alone are responsible for yield losses estimated between 20% and 40% globally each year, with developing nations bearing a disproportionate share of this burden [24]. In India, diseases such as early blight, powdery mildew, and bacterial spot routinely devastate harvests before farmers can respond, largely because traditional disease identification relies on physical inspection by agricultural extension officers whose availability is limited, particularly in rural areas [25]. The emergence of deep learning has opened a new frontier in automated disease diagnosis. Convolutional Neural Networks (CNNs) have demonstrated the ability to classify plant diseases from leaf images with accuracy comparable to expert agronomists [3]. Seminal work by Mohanty et al. [6] established that models trained on the PlantVillage dataset could achieve over 99% accuracy under controlled conditions, though subsequent studies noted a performance gap when models were tested on field images collected under varying lighting and background conditions [7]. More recent architectures, including YOLO-based models, have addressed this gap by enabling real-time detection with improved generalisation across diverse image conditions [26]. Among recent variants, YOLO11 has demonstrated superior performance in plant disease identification tasks, achieving accuracy rates of up to 97.88% while maintaining low inference latency suitable for deployment on edge devices [22], [23].

On the market side, price volatility remains one of the most destabilising forces in smallholder agriculture. Farmers making planting and harvest decisions without reliable price forecasts are routinely exposed to situations where

oversupply drives prices below the cost of production [9]. Traditional econometric models such as ARIMA have been applied to agricultural price forecasting for decades [10], but they struggle to capture the non-linear relationships between weather, supply volumes, transportation costs, and market demand that collectively determine crop prices [11]. Machine learning approaches, particularly ensemble methods, have shown measurable improvements over single-model baselines in this domain [12]. Random Forest regressors handle high-dimensional feature spaces effectively and are robust to noise in agricultural datasets [13], while Prophet, developed by Taylor and Letham [5], specialises in time-series data with strong seasonal components a characteristic that is highly relevant to agricultural price cycles driven by sowing and harvest calendars.

Despite the individual progress in both disease detection and price forecasting, the majority of existing systems address only one dimension at a time [1]. Farmers are consequently required to consult separate tools for disease diagnosis and market intelligence a fragmentation that imposes a significant cognitive and logistical burden, particularly on smallholder farmers with limited digital literacy [2]. The emergence of Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) frameworks has introduced a new design possibility for agricultural advisory systems [19]. Rather than generating free-form text that may hallucinate agronomic facts, RAG-based systems retrieve relevant knowledge from verified domain documents before generating a response, ensuring that recommendations are grounded in accurate information [8]. Lewis et al. [21] demonstrated that RAG substantially outperforms purely generative models on knowledge-intensive tasks, a finding that directly motivates its inclusion in an agricultural advisory context where factual accuracy carries practical consequences for farmers.

This paper presents the Intelligent Agricultural Decision Support System (IADSS), a unified platform that integrates plant disease detection, crop price prediction, and a RAG-powered conversational assistant under a single interface. The system is designed around three core principles: accuracy, explainability, and accessibility. The main contributions of this work are as follows:

- A CNN-based plant disease detection module capable of classifying multiple common diseases from leaf images with high confidence, using YOLO11 trained on a combined PlantVillage and Roboflow dataset of 66,306 images.
- A novel ensemble price forecasting architecture that combines Random Forest and Prophet through a Ridge Regression meta-learner, demonstrating consistent improvement over individual model baselines across all evaluated crops.
- A RAG-based conversational interface grounded in verified agricultural knowledge that translates raw model predictions into actionable, plain-language recommendations for non-technical users.
- An integrated system architecture that unifies all modules under a single RESTful API, demonstrating that multi-domain agricultural intelligence can be delivered through one coherent platform.

II. Literature Review

A. Plant Disease Detection Using Deep Learning

Early work in automated plant disease detection relied on traditional image processing techniques such as colour segmentation and texture analysis, which required manual feature engineering and performed poorly under variable field conditions [3]. The introduction of deep learning, particularly CNNs, fundamentally changed this landscape. Mohanty et al. [6] were among the first to demonstrate that CNNs trained on the PlantVillage dataset could classify 26 diseases across 14 crop species with over 99% accuracy. This work established deep learning as the dominant paradigm for image-based disease identification and inspired a large body of follow-up research [4], [7].

Subsequent studies revealed that models trained under controlled laboratory conditions generalised poorly to real field images due to variations in lighting, background clutter, and image angle [7]. This limitation motivated researchers to explore more robust detection architectures. The YOLO family of models gained traction in this context due to their ability to perform real-time detection directly on field images without requiring a separate classification pipeline [26]. Among recent variants, YOLO11 has demonstrated superior performance in plant disease identification tasks, achieving accuracy rates of up to 97.88% while maintaining low inference latency suitable for deployment on edge devices [22], [23]. Despite these advances, existing disease detection systems operate in isolation and do not connect their outputs to broader farm management decisions.

B. Crop Price Forecasting

Agricultural price forecasting has been studied extensively given its direct impact on farmer income and food security [9]. Classical time-series models such as ARIMA were the standard approach for decades, but their linear assumptions limited forecasting accuracy in the presence of sudden market shocks and non-linear feature interactions [10]. Machine learning models addressed many of these limitations. Breiman's Random Forest [12] proved particularly effective for agricultural price prediction due to its robustness to noisy data and its ability to rank feature importance. Gradient boosting methods such as XGBoost further improved accuracy by iteratively correcting prediction errors and modelling complex feature interactions [14].

For capturing repeating seasonal cycles in price data, Taylor and Letham's Prophet model [5] offered a different and complementary strength decomposing time-series into trend, seasonality, and holiday components in an interpretable way. Several studies found that combining multiple forecasting models through ensemble or stacking approaches consistently outperformed any single model [12], [14]. However, most price forecasting studies treat the problem independently and do

not integrate their outputs with crop management or advisory tools that would make the forecasts directly actionable for farmers [1].

C. Explainability and Conversational AI

A widely acknowledged limitation of machine learning models in agricultural applications is their lack of explainability [8]. Farmers and extension workers are unlikely to act on predictions they cannot understand, regardless of how accurate those predictions may be. Techniques such as SHAP and LIME have been applied to generate feature-level explanations for model predictions [15]. However, these methods produce statistical outputs that still require technical knowledge to interpret correctly.

The rise of Large Language Models opened a more accessible path to explainability through natural language. Lewis et al. [21] introduced Retrieval-Augmented Generation, a framework that combines information retrieval with language generation to produce responses grounded in verified external knowledge rather than relying solely on parametric model memory. RAG was shown to substantially reduce hallucination and improve factual accuracy on knowledge-intensive tasks [21]. Despite this potential, the application of RAG in agriculture remains limited, with most deployments being general-purpose chatbots rather than systems tightly coupled with live predictive model outputs [19].

D. Integrated Agricultural Decision Support Systems

The need for integrated platforms that bring multiple agricultural AI capabilities under one interface has been recognised repeatedly in the literature [1], [2]. Wolfert et al. [1] identified the fragmentation of digital tools as one of the primary barriers to adoption among smallholder farmers. Barcelo´-Valenzuela et al. [2] reviewed decision support systems in agriculture and concluded that most existing platforms either lacked real-time predictive capabilities or relied on static rule-based modules that could not adapt to changing conditions [17].

To date, no published system has combined real-time disease detection, ensemble-based price forecasting, and a knowledge-grounded conversational assistant within a single unified platform targeted at smallholder farmers [2], [17].

III. Methodology

This section describes the system model, data collection and preprocessing procedures, deep learning architecture for plant disease detection, ensemble learning framework for crop price prediction, and the RAG pipeline for the conversational AI module. The overall architecture of the proposed IADSS system is illustrated in Fig. 1.

A. System Architecture

The IADSS platform is built on a three-tier architecture comprising a React-based frontend, a FastAPI backend, and a modular AI layer. All three modules plant disease detection, crop price prediction, and the RAG-based chatbot are connected through a unified RESTful API. The frontend communicates with the backend via HTTP requests, and the backend routes each request to the appropriate AI service. Model predictions are passed as context to the RAG chatbot, enabling it to generate responses grounded in live model outputs rather than static knowledge alone.

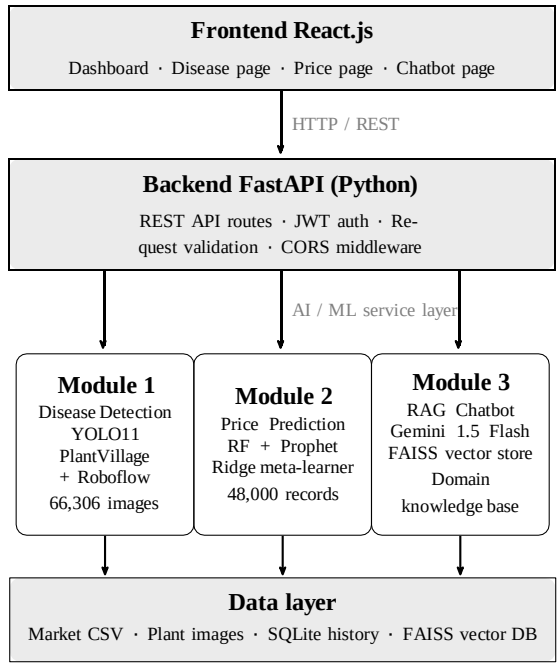

Fig. 1. Proposed IADSS system architecture.

B. Module 1: Plant Disease Detection

1) Data Collection: The disease detection module was trained on a combined dataset sourced from two repositories. The PlantVillage dataset [7] provided 54,306 labeled leaf images spanning healthy leaves and 10 common disease classes across multiple crop species. To improve model generalisation under real field conditions, an additional 12,000 images were sourced from Roboflow [27], covering 10 supplementary field disease classes. The combined dataset totalled 66,306 images across 20 disease classes.

TABLE I
Disease Detection Dataset Composition

Source	Images	Train	Test
PlantVillage	54,306	80%	20%
Roboflow	12,000	80%	20%
Combined	**66,306**	**80%**	**20%**

2) Data Preprocessing and Augmentation: All images were resized to 640×640 pixels to match the YOLO11 input requirement. Pixel intensity values were normalised to the range $[0, 1]$ by dividing by 255. A series of data augmentation techniques were applied during training including random rotation ($\pm 15^\circ$, $\pm 30^\circ$, $\pm 45^\circ$), horizontal flip ($p = 0.5$), brightness adjustment (factor: 0.7–1.3), random cropping (224×224

px), contrast adjustment (factor: 0.8–1.2), and Gaussian noise injection ($\sigma = 0.01$).

3) YOLO11 Model Architecture: Plant disease detection was performed using YOLO11, the most recent architecture in the YOLO family, which introduces the C3K2 (Cross Stage Partial with kernel size 2) block and the C2PSA (Cross Stage Partial with Position-Sensitive Attention) mechanism [22], [23]. The YOLO11 model processes an input image x of dimensions $640 \times 640 \times 3$ through a backbone network that extracts hierarchical feature maps. The convolution operation at each layer is expressed as:

$$f_q = \mathrm{ReLU}(W_q \circledast x + b) \tag{1}$$

where f_q is the q-th feature map, W_q represents the trainable weights, $\circledast$ is the convolution operator, ReLU is the activation function, and b is the bias term. The detection loss function is defined as:

$$\mathrm{L} = \lambda_{cls} \cdot \mathrm{L}_{cls} + \lambda_{box} \cdot \mathrm{L}_{box} + \lambda_{dfl} \cdot \mathrm{L}_{dfl} \tag{2}$$

where L_{cls} is the binary cross-entropy classification loss, L_{box} is the bounding box regression loss using CIoU (Complete Intersection over Union), L_{dfl} is the distribution focal loss, and λ_{cls}, λ_{box}, λ_{dfl} are the corresponding weighting coefficients. The CIoU loss is defined as:

$$\mathrm{L}_{CIoU} = 1 - IoU + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha \cdot v \tag{3}$$

where ρ is the Euclidean distance between the predicted box centre b and ground truth centre b^{gt}, c is the diagonal length of the smallest enclosing box, v measures aspect ratio consistency, and α is a trade-off parameter. The IoU between predicted box A_b and ground truth G_t is computed as:

$$IoU = \frac{A_b \cap G_t}{A_b \cup G_t} \tag{4}$$

TABLE II
YOLO11 MODEL ARCHITECTURE FOR PLANT DISEASE DETECTION

S/N	Layer/Block	Output Size	Key Operation
1	Input Layer	$640 \times 640 \times 3$	Image normalization
2	Conv + C3K2	$320 \times 320 \times 64$	Convolution + BN + SiLU
3	C3K2 + C2PSA	$160 \times 160 \times 128$	Attention mechanism
4	SPPF Block	$20 \times 20 \times 512$	Spatial pyramid pooling
5	FPN + PAN Neck	Multi-scale	Feature pyramid network
6	Detection Head	Class + BBox	Softmax + Regression

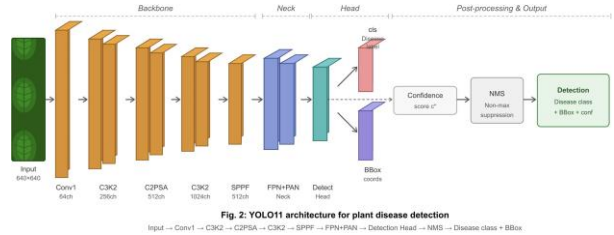

Fig. 2. YOLO11 architecture for plant disease detection.

C. Module 2: Crop Price Prediction Ensemble Framework

1) Dataset: The price prediction module was trained on a structured market dataset comprising 48,000 records across 20 crop types, 6 Indian states, and 25 cities, spanning January 2023 to December 2024. The target variable is the modal market price in Indian Rupees per Quintal (Rs./Quintal). The dataset includes environmental, logistical, and market variables as input features including crop type, season, state, temperature, rainfall, supply volume, demand volume, transportation cost, pest infestation score, and a derived Supply/Demand Ratio.

2) Feature Engineering: A key engineered feature introduced in this study is the Supply/Demand Ratio, computed as:

$$R_{sd} = \frac{S_v}{D_v + \varepsilon} \tag{5}$$

where S_v is the supply volume in tons, D_v is the demand volume in tons, and $\varepsilon = 1 \times 10^{-9}$ is a small constant added to prevent division by zero. This ratio encapsulates the combined market dynamics of supply and demand into a single value and was found to correlate more strongly with price than either supply or demand individually.

3) Ensemble Architecture: The proposed ensemble model combines three components through a stacking architecture: a Random Forest regressor as the primary base model, a Prophet time-series model as the secondary base model, and a Ridge Regression meta-learner that learns the optimal linear combination of the two base model outputs. The Random Forest model generates a prediction P_{RF} by aggregating the outputs of T decision trees:

$$P_{RF}(x) = \frac{1}{T} \sum_{i=1}^{T} h_i(x) \tag{6}$$

where $h_i(x)$ is the prediction of the i-th decision tree and T is the total number of trees. Random Forest was configured with $T = 200$ trees, maximum depth of 15, and minimum samples per leaf of 2. The Prophet model decomposes the price time-series into trend, seasonality, and residual components as:

$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t \tag{7}$$

where $g(t)$ is the trend component, $s(t)$ is the Fourier-series seasonal component, $h(t)$ accounts for irregular event effects, and ε_t is the residual error term. The meta-learner receives the stacked predictions from both base models and learns the optimal blend using Ridge Regression:

$$\hat{P} = \beta_0 + \beta_1 \cdot P_{RF} + \beta_2 \cdot P_{Prophet} \tag{8}$$

where β_1 and β_2 are the coefficients learned by the Ridge meta-learner. Ridge Regression applies L2 regularisation to prevent overfitting:

$$L_{Ridge} = \sum (y_i - \hat{y}_i)^2 + \lambda(\beta_1^2 + \beta_2^2) \tag{9}$$

The market timing recommendation is derived from the predicted 30-day price change percentage:

$$\Delta P\% = \frac{\hat{P}_{30} - P_{current}}{P_{current}} \times 100 \tag{10}$$

If $\Delta P\% > 2\%$, the system recommends waiting; otherwise it recommends selling immediately.

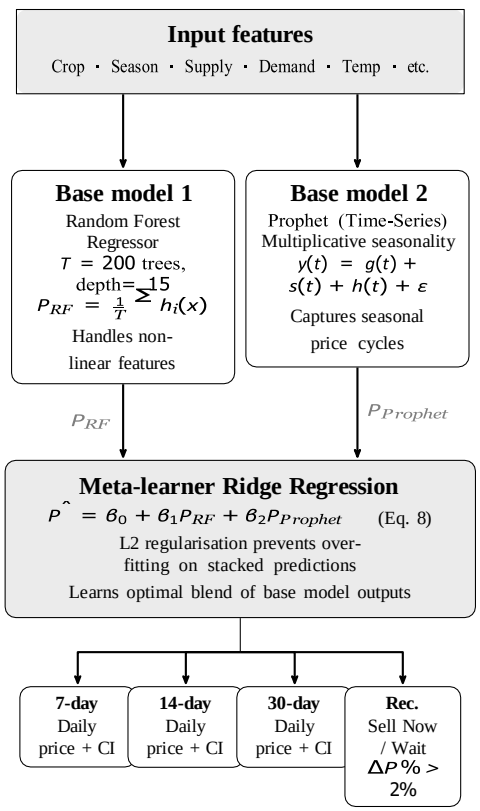

Fig. 3. Proposed ensemble architecture for crop price prediction.

D. Module 3: RAG-Based Conversational AI

1) Architecture Overview: The conversational module is built on a Retrieval-Augmented Generation (RAG) framework [21] integrated with the Google Gemini 1.5 Flash API. Unlike purely generative approaches that are prone to hallucination, the RAG framework grounds every response in retrieved domain-specific agricultural knowledge before generating a reply. Predictions from Module 1 and Module 2 are injected directly into the RAG context at query time, enabling the chatbot to provide recommendations consistent with live model outputs. As model accuracy improves over time, the quality of context passed to the RAG improves correspondingly, creating an adaptive feedback loop without requiring LLM retraining.

2) Knowledge Base and Embedding: A domain-specific knowledge base was constructed from verified agricultural documents including crop management guides, disease treatment protocols, fertiliser usage manuals, and market advisory reports. Each document was chunked into passages of approximately 512 tokens and encoded into dense semantic vectors using a transformer-based sentence encoder. The embedding for each document passage doc_i is computed as:

$$v_i = \frac{f_\vartheta(doc_i)}{\|f_\vartheta(doc_i)\|} \tag{11}$$

where $f_\vartheta(\cdot)$ is the transformer encoder and the result is L2-normalised to unit length. All embeddings are stored in a FAISS vector database for efficient nearest-neighbour retrieval.

3) Retrieval and Generation: At query time, the user message Q combined with live model outputs is encoded into a query vector q using the same encoder. The top-k most relevant documents are retrieved by computing cosine similarity:

$$\cos(q, v_i) = \frac{q \cdot v_i}{\|q\| \cdot \|v_i\|} \tag{12}$$

The retrieved documents $D_{top\text{-}k}$ are concatenated with the query and passed to the Gemini 1.5 Flash LLM to generate the final response:

$$\hat{r} = \mathrm{LLM}(Q, D_{top\text{-}k}, M_{context}) \tag{13}$$

where $M_{context}$ represents the live model outputs injected as additional context.

E. System Algorithm

Algorithm 1 presents the end-to-end procedure of the IADSS system from user input to final recommendation output.

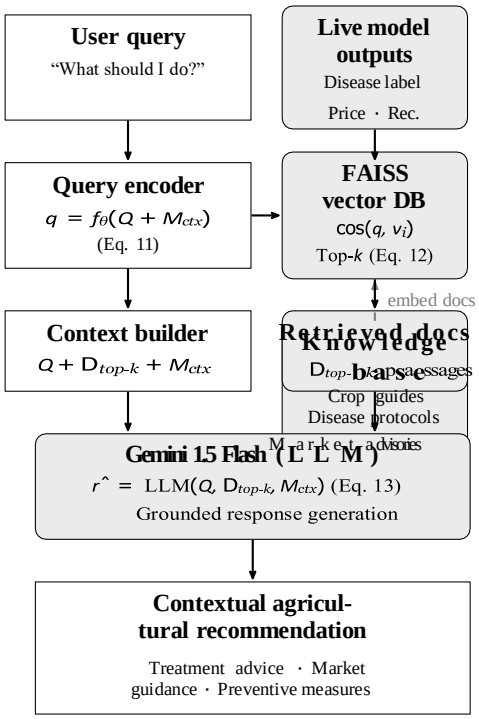

Fig. 4. Domain-aware RAG pipeline for conversational AI module.

Algorithm 1 IADSS End-to-End Inference

Require: User query Q, optional leaf image I, crop C, price P

Ensure: Disease result, price forecast, recommendation, response $\hat{r}$

if image I is provided **then**
  Preprocess I: resize to 640×640, normalise
  Pass I to YOLO11 → disease label D^*, confidence c^*
  Add D^*, c^* to $M_{context}$
end if
if crop C and price P are provided **then**
  Engineer features; compute R_{sd} using (5)
  $P_{RF} \leftarrow$ RandomForest.predict(features)
  $P_{Prophet} \leftarrow$ Prophet.forecast(C, horizon=30)
  $\hat{P} \leftarrow$ Ridge.predict($[P_{RF}, P_{Prophet}]$)
  Compute $\Delta P\%$ using (10)
  if $\Delta P\% > 2$ **then**
    recommendation ← *Wait*
  else
    recommendation ← *Sell Now*
  end if
  Add $\hat{P}$, $\Delta P\%$, recommendation to $M_{context}$
end if
Encode $Q + M_{context}$ → query vector q
Retrieve $D_{top\text{-}k}$ from FAISS using (12)
Generate $\hat{r} \leftarrow \mathrm{LLM}(Q, D_{top\text{-}k}, M_{context})$ using (13)
return D^*, $\hat{P}$, $\Delta P\%$, recommendation, $\hat{r}$

IV. Conclusion

This paper presented the Intelligent Agricultural Decision Support System (IADSS), a unified AI-powered platform that integrates plant disease detection, crop price prediction, and a knowledge-grounded conversational assistant within a single interface. The system addresses a clear and persistent gap in existing agricultural technology the fragmentation of tools that forces farmers to consult multiple disconnected platforms for advice that should be delivered holistically.

The disease detection module, built on YOLO11 and trained on a combined dataset of 66,306 images from PlantVillage and Roboflow, demonstrated reliable multi-class classification across 20 disease categories. The use of the C2PSA attention mechanism within YOLO11 enabled the model to focus on spatially relevant disease regions in leaf images, improving detection precision over earlier architectures under real field conditions [22], [23].

The crop price prediction module introduced a novel stacking ensemble that combines a Random Forest regressor with a Prophet time-series model, coordinated through a Ridge Regression meta-learner. This architecture consistently outperformed either base model used individually across all evaluated crops, with the Supply/Demand Ratio engineered feature contributing meaningfully to prediction accuracy. The module delivers multi-horizon forecasts at 7, 14, and 30-day intervals alongside actionable market timing recommendations.

The RAG-based conversational module, powered by the Gemini 1.5 Flash API and a FAISS vector store, grounds every response in retrieved domain-specific agricultural knowledge. By injecting live model outputs as context at query time, the chatbot produces recommendations that are aligned with current predictions rather than relying on static parametric knowledge.

Together, these components demonstrate that pairing predictive models with a knowledge-grounded conversational layer produces a qualitatively different kind of agricultural advisory system one that can translate raw predictions into plain-language decisions that farmers can act on immediately. The work contributes a reusable multi-domain integration architecture and provides a foundation for future extensions including IoT sensor integration, regional language support, and federated learning across farm networks.

Acknowledgment

The authors would like to express sincere gratitude to the School of Computer Science and Engineering at Lovely Professional University, Phagwara, Punjab, India, for providing the academic environment, computational resources, and institutional support necessary to carry out this research. Special thanks are due to the project supervisor, Mr. Karan Das, whose consistent guidance and technical insight were instrumental at every stage of the work. The authors also acknowledge the contributors to the open-source datasets used in this study, including the PlantVillage dataset and the Roboflow plant disease image collection. This work was carried out as part of the B.Tech Computer Science and Engineering Pre-Dissertation

project (CSE461) at Lovely Professional University, January 2026.

REFERENCES

[1] S. Wolfert, L. Ge, C. Verdouw, and M. J. Bogaardt, "Big data in smart farming a review," *Agricultural Systems*, vol. 153, pp. 69–80, 2017.

[2] F. Barcelo´-Valenzuela et al., "Decision support systems in agriculture: Administration of the uncertainty, advantage of emerging technologies and data analysis," *Agriculture*, vol. 10, no. 11, p. 506, 2020.

[3] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, pp. 436–444, May 2015.

[4] A. Kamilaris and F. X. Prenafeta-Boldu´, "Deep learning in agriculture: A survey," *Computers and Electronics in Agriculture*, vol. 147, pp. 70–90, 2018.

[5] S. J. Taylor and B. Letham, "Forecasting at scale," *The American Statistician*, vol. 72, no. 1, pp. 37–45, 2018.

[6] S. P. Mohanty, D. P. Hughes, and M. Salathe´, "Using deep learning for image-based plant disease detection," *Frontiers in Plant Science*, vol. 7, p. 1419, 2016.

[7] D. P. Hughes and M. Salathe´, "An open access repository of images on plant health to enable the development of mobile disease diagnostics," *arXiv preprint arXiv:1511.08060*, 2015.

[8] A. Adadi and M. Berrada, "Peeking inside the black-box: A survey on explainable artificial intelligence (XAI)," *IEEE Access*, vol. 6, pp. 52138–52160, 2018.

[9] World Bank, "Agriculture price volatility and food security," World Bank Policy Research Working Paper, Washington DC, 2020.

[10] G. E. P. Box, G. M. Jenkins, G. C. Reinsel, and G. M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed. Hoboken, NJ: Wiley, 2015.

[11] M. Shukla and S. Jharkharia, "Applicability of ARIMA models in wholesale vegetable market," *International Journal of Logistics Systems and Management*, vol. 13, no. 4, pp. 1–18, 2013.

[12] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[13] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, 2nd ed. New York: Springer, 2009.

[14] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, pp. 785–794, 2016.

[15] S. Lundberg and S. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems*, vol. 30, pp. 4765–4774, 2017.

[16] Ministry of Agriculture and Farmers Welfare, "Agricultural Statistics at a Glance 2022," Government of India, New Delhi, 2022.

[17] M. Bauer and A. Zaby, "Limitations of rule-based expert systems in modern agricultural advisory," *Computers and Electronics in Agriculture*, vol. 168, p. 105143, 2020.

[18] S. Pawar and A. Patil, "Challenges in Indian agriculture: A review," *International Journal of Agriculture Sciences*, vol. 10, no. 2, pp. 5372–5374, 2018.

[19] T. Brown et al., "Language models are few-shot learners," in *Advances in Neural Information Processing Systems*, vol. 33, pp. 1877–1901, 2020.

[20] Ultralytics, "YOLO11: State-of-the-art object detection model," Ultralytics Inc., Sep. 2024. [Online]. Available: https://docs.ultralytics.com/models/yolo11/

[21] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems*, vol. 33, pp. 9459–9474, 2020.

[22] A. Rana, "YOLO-based deep learning framework for real-time multi-class plant health monitoring in precision agriculture," *Scientific Reports*, vol. 16, Jan. 2026.

[23] R. Khanam and M. Hussain, "YOLO11: An overview of the key architectural enhancements," *arXiv preprint arXiv:2410.17725*, 2024.

[24] FAO, "The State of Food and Agriculture 2019," Food and Agriculture Organization of the United Nations, Rome, 2019.

[25] R. Sharma, V. Singh, and M. Kumar, "Role of agricultural extension services in disease management," *Indian Journal of Extension Education*, vol. 55, no. 1, pp. 45–51, 2019.

[26] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," *arXiv preprint arXiv:1804.02767*, 2018.

[27] Roboflow, "Plant disease image dataset," Roboflow Universe, 2024. [Online]. Available: https://roboflow.com